Assignment 7: Random Forest Feature Selection

Project Overview

This project demonstrates feature selection using a Random Forest classifier on the Wine dataset. It identifies the most important features, selects the top five, and compares the performance of models trained with all features and the selected features.

Objectives

- Train a baseline Random Forest model using all available features.
- Calculate and visualize feature importance.
- Select the top five most important features.
- Retrain the model using only the selected features.
- Compare model accuracy, simplicity, and interpretability.

Dataset

- **Dataset:** Wine Dataset
- **Source:** Scikit-learn built-in dataset
- **Total Samples:** 178
- **Input Features:** 13
- **Target Classes:** 3

The dataset is loaded directly using sklearn.datasets.load_wine().

Technologies Used

- Python
- Google Colab
- Pandas
- Matplotlib
- Scikit-learn

Project Workflow

1. Import the required Python libraries.
2. Load and explore the Wine dataset.
3. Split the dataset into training and testing sets.

4. Train a baseline Random Forest classifier using all 13 features.
5. Calculate and visualize feature importance.
6. Select the five most important features.
7. Train a second Random Forest classifier using the selected features.
8. Compare the accuracy of both models.
9. Analyze the impact of feature selection.

Model Evaluation

The two models are evaluated using:

- Accuracy score
- Precision
- Recall
- F1-score
- Classification report

A bar chart is used to compare the accuracy of the baseline model and the top-five-features model.

Results

The notebook generates:

- Feature importance ranking
- Feature importance visualization
- List of the top five selected features
- Classification reports for both models
- Model accuracy comparison chart

The actual results are calculated when the notebook is executed.

Key Findings

Feature selection reduces the number of input features from 13 to 5, making the model easier to analyze and reducing the amount of input data required.

The effect on accuracy depends on how much useful information the selected features retain. The comparison helps determine whether a smaller feature set can maintain competitive predictive performance.

How to Run the Project

1. Open Google Colab.
2. Upload Assignment_7_Random_Forest_Feature_Selection.ipynb.
3. Run each code cell from top to bottom.
4. Review the feature importance graph and model comparison results.

No manual dataset download is required.

Conclusion

This project demonstrates how Random Forest feature importance can be used for feature selection. By comparing models trained with all features and the top five features, it evaluates the trade-off between predictive accuracy, simplicity, and interpretability.

Author

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